

Course Code: CS1103 Name of the Programme: B.Tech(H) Name of the Course: Operating Systems Semester: II		
Course Credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	2-0-2	60
Prerequisite: - Sem 1 CS1004: Programming in C - Sem 1 CS1101: Digital Systems and Computer Architecture		
Course Objectives:		
A. Understand the core concepts, architecture, and services of modern operating systems.		
B. Explore process management, scheduling, and multithreading techniques.		
C. Learn synchronization methods and deadlock handling strategies.		
D. Examine memory management, file systems, storage, and system security.		

Syllabus:

Module – 1: Introduction to Operating Systems	No. of hours
	6
Operating Systems –Definition, OS Types, OS Functionalities, OS Services, Abstract view of OS-Computer System Structure, Boot Process, System Calls, Virtualization - Virtual Machines.	
Module – 2: Process Management	No. of hours
	6
Process Concepts, Inter-process communication –Threads –Multithreading, Process Scheduling, First-Come, First-Served (FCFS), Shortest Job First, Round Robin (RR), Priority Scheduling.	
Module – 3: Synchronization and Deadlocks	No. of hours
	6
Synchronization –Semaphores, Mutex Locks –Monitors, –Deadlocks –Methods for Handling Deadlocks, Ignorance, Avoidance, Prevention, Detection and Recovery.	

Module – 4: Memory Management	No. of hours
	6
Memory Management Strategies –Contiguous and Non-Contiguous allocation –Paging - Page replacement FCFS, LRU, Optimal, LFU Algorithms–Virtual Memory Management – File System Design and Implementation	

Module - 5	No. of hours
	6
Mass Storage Structure –Disk Scheduling and Management, FCFS, SSTF, and SCAN, C-SCAN, LC C-LOOK -I/O Systems- Overview of System Protection and Security.	

Course Outcomes:
a. Understand the structural components and core functionalities of an operating system.
b. Apply process management techniques and scheduling algorithms to ensure efficient execution resource allocation.
c. Apply synchronization techniques and design solutions to prevent or resolve deadlocks in concurrent systems.
d. Illustrate the memory management strategies, principles of file system design, and security mechanisms for efficient performance, resource utilization, and reliable data handling in operating systems.

Text Books	
1	A. Silberschatz, P. B. Galvin, and G. Gagne, <i>Operating System Concepts</i> , 10th ed. Hoboken, NJ, USA: John Wiley & Sons, 2023, ISBN: 935746056X.
2	W. Stallings, <i>Operating Systems: Internals and Design Principles</i> , 9th ed. Boston, MA, USA: Pearson, Mar. 2017, ISBN: 0134670957.

Reference Books	
1	A. S. Tanenbaum and H. Bos, <i>Modern Operating Systems</i> , 4th ed. Boston, MA, USA: Pearson, Mar. 2014, ISBN: 978-0-13-359162-0.
2	R. Petersen, <i>Linux: The Complete Reference</i> , 6th ed. New Delhi, India: Tata McGraw-Hill, 2008, ISBN: 007149247X.
3	M. Kerrisk, <i>The Linux Programming Interface</i> , 1st ed. San Francisco, CA, USA: No Starch Press, 2010, ISBN: 978-1593272203.

Lab Programs	
1	1.1) Exploring basic Linux commands and Shell programs. 1.2) Execute the C program by passing parameters as Command-Line arguments. i) Add numbers ii) Find the average of the numbers iii) Find the minimum from the given numbers
2	Execute C programs to implement Process system calls like fork(), exec(), wait() and exit().
3	Execute C programs to simulate CPU scheduling algorithms to find turnaround and waiting time. 3.1) First Come First Served (FCFS) algorithm 3.2) Shortest Job First (SJF) algorithm 3.3) Round Robin (RR) algorithm 3.4) Priority Scheduling algorithm
4	Execute C programs for Thread Creation and Management using the POSIX pthread library.
5	Execute a C Program for IPC using the pipe () function.
6	Execute a C program to simulate the producer-consumer problem using semaphores.
7	Execute the Banker's Algorithm for deadlock avoidance using the C program.
8	Execute C programs to simulate Page Replacement Algorithms. 8.1) First In First Out (FIFO) algorithm 8.2) Least Recently Used (LRU) algorithm
9	Execute C programs to display disk scheduling algorithms. 9.1) First Come First Serve (FCFS) algorithm 9.2) Shortest Seek Time First (SSFT) algorithm 9.3) SCAN algorithm

